

Lesson 18: The Unit Circle & Exact Values — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Exact value of $\sin 30^\circ$.
2. Exact value of $\cos 60^\circ$.
3. Exact value of $\tan 45^\circ$.
4. Exact value of $\sin 90^\circ$.
5. Exact value of $\cos 0^\circ$.
6. Exact value of $\sin 45^\circ$.

Section 2 — Core practice

7. Exact value of $\cos 30^\circ$.
8. Exact value of $\tan 60^\circ$.
9. Exact value of $\sin 120^\circ$.
10. Exact value of $\cos 120^\circ$.
11. Exact value of $\cos 180^\circ$.

12. In which quadrant is only cosine positive?

Section 3 — Challenge & exam-style

13. Find the exact value of $\cos 150^\circ$.

14. Find the exact value of $\sin 210^\circ$.

15. Find the exact value of $\tan 225^\circ$.

16. Given $\cos \theta = 4/5$ and θ acute, find $\sin \theta$.